

PitStop

Master System Documentation

Complete reference: customer app, owner panels, Super Admin, database, auth, booking, and deployment.

التوثيق الشامل — عمل، صاحب محل، مدير فرع، Super Admin، قاعدة البيانات، والحجز

[Expo 54](#)[React Native](#)[Supabase](#)[Super Admin](#)[AR / EN](#)[PitStop 2.0](#)

Project: car-care-eg · Version 1.0.0 · Updated with Super Admin workflow

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1. Architecture Overview

نظرة عامة على البنية

Stack

Layer	Technology
Mobile / Web UI	Expo Router ~54, React Native, expo-linear-gradient, expo-image
State	React Context (auth, i18n, theme, shop catalog)
Local storage	AsyncStorage (bookings, extras, notifications, wash branches cache)
Legacy garage	SQLite via expo-sqlite (catalog, odometer — hidden tabs)
Backend	Supabase Auth + Postgres (optional; app works offline-first)
Push	Expo Notifications → shop_push_tokens (native only)
AI	OpenAI GPT-4o-mini (Assistant tab)

Provider tree (startup)

ThemePreferenceProvider → I18nProvider → ShopCatalogProvider
→ CustomerAuthProvider → AuthProvider → ShopAuthProvider → AppBootstrap

Data flow (hybrid)

Customer action (book, order)
→ lib/booking/storage.ts or partsStorage.ts
→ Supabase insert (if configured) + AsyncStorage cache
→ Owner notification (local + optional Expo push)
→ Owner approves/declines → customer in-app notification

Shop type enum

maintenance · wash · parts · accessories · winch

2. App Routes & Screens

المسارات والشاشات

Root stack (app/_layout.tsx)

Route	File	Purpose
/welcome	welcome.tsx	Customer + owner login, merchant registration, guest, language
/(tabs)/*	(tabs)/_layout.tsx	Main tab navigator
/service/[type]	service/[type]/index.tsx	Pick area (Cairo/Giza) for service type
/service/[type]/[areald]	service/[type]/[areald].tsx	Shops in area → profile or parts shop
/shop-profile/[shopId]	shop-profile/[shopId].tsx	Public profile: gallery, hours, services, reviews, book
/book/[shopId]	book/[shopId].tsx	Booking form: date, slots, car, phone, notes
/nearby/[type]	nearby/[type].tsx	GPS discovery + filters (top rated, price, open now...)
/parts-shop/[shopId]	parts-shop/[shopId].tsx	Parts/accessories cart + checkout
/auth-required	auth-required.tsx	Gate before booking/purchase for guests
/settings/vehicles	settings/vehicles.tsx	Customer vehicle list
/shop/wash-owner-hub	shop/wash-owner-hub.tsx	Wash hub: notifications, orders, history
/admin	admin/index.tsx	Super Admin panel (dashboard, pending, merchants)
/reset-password	reset-password.tsx	Supabase password reset deep link
/login, /verify	login.tsx, verify.tsx	Legacy phone OTP (garage prototype)
/car/[id], /add-car	car/, add-car.tsx	Legacy Egypt garage (SQLite odometer)

Customer tabs (app/(tabs)/)

Tab	Visible	Purpose
index (Home)	Customer/guest	Service cards, offers, next booking
bookings	Customer/guest	My bookings, cancel, history
favorites	Logged-in customer	Saved shops
assistant	Customer/guest	Driver Network + AI chat
settings	Customer/guest	Language, theme, vehicles, support, logout
shop	Shop session	Owner login or owner dashboard
catalog, alerts	Hidden	Legacy garage features

Navigation rules (AppBootstrap.tsx)

- Public paths: /welcome, /reset-password, /auth-required, /login, /verify
- **Admin session** → redirect to /admin
- **Pending owner** → stay on /welcome?pending=1 (no shop access)
- Approved shop session → redirect to /shop
- No session on protected route → /welcome

- Customer on /welcome after login → home or returnUrl deep link

3. Authentication & Roles

المصادقة والأدوار

Roles (`public.user_role`)

Role	Login	shop_id	branch_id	Purpose
customer	Yes	—	—	Books services, favorites, notifications
owner	Yes	Required	—	Owns brand, all branches, creates managers
branch_manager	Yes	Required	Required	Runs one branch day-to-day
pending_owner	Yes	Required	—	New merchant awaiting admin approval; <code>is_active=false</code>
admin	Yes	—	—	Super Admin → /admin panel

Employees are NOT auth users — stored in `branch_employees` only.

Shop visibility: `shops.is_active=false` hides shop from customer catalog until admin approves.

Customer auth (`CustomerAuthContext.tsx`)

- Supabase email/password: `signIn`, `signUp`, `resetPassword`
- Guest mode: local flag, no Supabase session
- Blocks shop staff emails from customer login
- Requires email confirmed for login
- Storage: `@pitstop/customer-session`, `@pitstop/guest-session`

Shop auth (`ShopAuthContext.tsx`)

- Resolves from `public.users` (`owner` | `branch_manager` | `admin` | `pending_owner`) or legacy `shops.owner_email`
- Flags: `isOwner`, `isBranchManager`, `isAdmin`, `isPendingOwner`
- `registerOwner()` — `signUp` + RPC `register_shop_owner`
- Login results: `ok` | `pending_approval` | `invalid_credentials` | ...
- Storage: `@pitstop/shop-session` (staff email)
- Branch manager invite uses `staffInviteLock` to prevent session hijack

Auth storage on web

`lib/supabase/authStorage.ts` uses `localStorage` on web for reliable Supabase session persistence.

Role hierarchy (updated)

```
Super Admin (platform)
├─ Approves / rejects pending_owner registrations
├─ Views platform stats (bookings, 12% fee)
└─ Sees all merchants + branches

Owner (approved)
├─ Creates branches
├─ Creates / links Branch Managers
└─ Sees all branches + reports

pending_owner (awaiting approval)
└─ Shop hidden from customers until admin accepts
```

Branch Manager (one branch)

- └─ Manages daily operations
- └─ Adds employees (no login)

Employees → roster only (branch_employees)

4. Customer Features

مميزات العميل

Home (`/(tabs)/index.tsx`)

- Service entry cards: maintenance, wash, parts, accessories
- Live promotional offers from shops
- Next upcoming booking snippet
- Car profile shortcut

Discovery

- **By area:** `/service/{type}` → pick Maadi, Nasr City, etc.
- **Nearby GPS:** `/nearby/{type}` with filters (top rated, price, distance, open now, favorites)
- **Favorites:** saved shops tab (login required)

Shop profile

- Gallery, bilingual name/address, phone, directions, share
- Services list with prices and duration
- Reviews (ratings, comments, likes, owner replies)
- Offers / coupons display
- More info section (owner-configured)
- Book appointment CTA → requires login (not guest)

My bookings

- List all bookings for customer phone/account
- Cancel pending/confirmed bookings
- Clear history
- In-app notifications for approval, decline, reminders

Driver Network + AI (`/(tabs)/assistant.tsx`)

- Community posts (local AsyncStorage)
- Embedded AI assistant (OpenAI — requires API key)

Settings

- Arabic / English toggle
- Light / dark theme
- Vehicle management link
- Support, privacy, sign out

5. Shop Types & Owner Panels

أنواع المحلات ولوحات التحكم

Panel routing (app/(tabs)/shop.tsx)

wash type → WashOwnerPanel.tsx (full CMS)
parts/accessories → Store owner UI (inventory + orders)
maintenance/winch → Generic owner UI (profile + bookings + PDF)

Generic owner panel (maintenance, winch)

Section	Features
Profile	Bilingual name, address, phone, email, more info; winch toggle (maintenance)
Gallery	Add/remove photos
Pricing	Default service price (EGP)
Schedule	Open/close, slot duration; Friday closed default
Offers	Bilingual promotional offers
Bookings	Active requests (confirm/cancel/done), history, date-range reports
PDF report	View report modal + save/share PDF
Notifications	Booking + parts order alerts

Store owner panel (parts, accessories)

Section	Features
Profile + gallery	Same as generic (no schedule/bookings)
Inventory	Add items: name, price, stock, image; ±1 stock
Orders	Confirm / cancel / mark shipped; 12% platform fee on orders

Wash owner panel (WashOwnerPanel.tsx) — PitStop 2.0

Section	Owner	Branch Manager
Dashboard stats	All branches	Own branch
Branch switcher	Yes	No
Create branch	Yes	No
Profile, gallery, base price	Per branch	Own branch
Services CRUD	Per branch	Own branch
Weekly hours + vacation	Per branch	Own branch
Coupons	Per branch	Own branch
Bookings workflow	All branches	Own branch
Employees roster	All branches	Own branch (add/edit)
Create branch manager	Yes	No
Reviews moderation	Yes	Yes (own branch)

PDF reports	All	Own branch
-------------	-----	------------

Service categories (wash)

exterior_wash · interior_cleaning · full_package · detailing · engine_cleaning · custom

Wash owner hub (/shop/wash-owner-hub)

Tab	Purpose
notifications	Wash notification inbox
orders	Approve/reject/update booking statuses
history	Past bookings; clear history

6. Booking Flow

مسار الحجز من البداية للنهاية

End-to-end workflow

```
Home / Nearby / Favorites
→ Shop profile (/shop-profile/[shopId])
→ [Guest?] → /auth-required → /welcome login
→ Book (/book/[shopId]?serviceId=...)
→ Select date + time slot + car + phone + notes
→ createBooking() → status: pending
→ Owner notification (+ optional push)
→ Owner: confirm → confirmed | decline → cancelled
→ Customer notification + reminders (60 min, 30 min before)
→ Owner: in_progress → done (or no_show)
```

Booking statuses

pending → confirmed → in_progress → done | cancelled | no_show

Pricing (lib/booking/storage.ts)

Shop type	Default price (EGP)
maintenance	650
wash	220
winch	500
other	420

Platform fee: **12%** of service price (shown in owner PDF reports).

Time slots

Built by `buildSlotsForShopDate()` from owner weekly hours + existing bookings. Submit disabled until a slot is selected.

Persistence

- Primary: Supabase `bookings` table (when configured)
- Cache: `AsyncStorage @pitstop/bookings/v1`
- On Supabase insert failure → error shown to customer (RLS fix: step4 SQL)

7. Parts & Store Orders

طلبات قطع الغيار والإكسسوارات

```
/parts-shop/[shopId]
  → Browse inventory (store table)
  → Cart + shipping address + phone
  → createPartsOrder()
  → Owner notification + push
  → Owner: confirm / cancel / ship
  → Customer notification
```

Order statuses: pending → confirmed → shipped | cancelled

8. Database Schema

Entity diagram

```
auth.users (Supabase Auth)
└─ public.users (role, shop_id, branch_id)

public.shops (brand)
├─ shop_branches (locations, hours, gallery, status)
│   └─ branch_services
│       └─ branch_employees (no login)
├─ bookings (+ branch_id, service_id)
├─ store (parts/accessories inventory)
└─ parts_orders + parts_order_items

shop_push_tokens · garage_snapshots (legacy) · areas
```

Core tables

Table	Purpose
areas	Cairo/Giza districts (Maadi, Nasr City, Mohandessin, October, Heliopolis)
shops	Brand catalog entry; legacy owner_email
users	App identity: customer, owner, branch_manager, admin
shop_branches	Physical branches with profile, weekly_hours, vacation_mode
branch_employees	Staff roster (full_name, phone, job_title, notes)
branch_services	Per-branch service menu
bookings	Appointments with status, pricing, branch_id
store	Parts/accessories SKUs with stock_qty
parts_orders	Customer store orders
shop_push_tokens	Expo push tokens per shop

RLS helpers

is_shop_owner · is_branch_manager · can_manage_shop · can_manage_branch

9. Supabase Setup & Migrations

إعداد Supabase وملفات ال SQL

Fresh project

1. supabase/schema.sql
2. supabase/seed.sql — 12 demo shops
3. supabase/seed-branches.sql — optional 2nd branch + employees

Existing project (PitStop 2.0 migration)

1. apply-pitstop-2.0-step1-enums.sql — run alone first
2. apply-pitstop-2.0-step2-main.sql — users, branches, employees
3. apply-pitstop-2.0-step3-branch-manager-rpc.sql — assign_branch_manager RPC
4. apply-pitstop-2.0-step4-bookings-rls.sql — restore booking insert policies
5. apply-pitstop-2.0-step5-admin-approval.sql — Super Admin, pending_owner, approve/reject RPCs
6. seed-test-data.sql — optional demo manager link

Other migration files

- migrate-roles-branches.sql — legacy → 2.0
- migrate-to-store.sql — spare_parts → store
- apply-pitstop-2.0-on-existing-db.sql — combined path

Step 5 RPCs (Super Admin)

RPC	Caller	Effect
register_shop_owner(...)	New merchant (authenticated)	Creates inactive shop + main branch; user → pending_owner
approve_shop_owner(user_id, shop_id)	Admin	user → owner, is_active=true; shop → is_active=true (live)
reject_shop_owner(user_id, shop_id)	Admin	Deletes shop; user → customer

Link owner after Auth user created

```
update public.users
set role = 'owner', shop_id = 'shop-wash-nile'
where email = 'wash@demo.com';
```

10. Key Library Modules

أهم ملفات /lib

Module	Role
lib/booking/storage.ts	Bookings CRUD, Supabase + AsyncStorage merge
lib/booking/catalogRepository.ts	Areas/shops from Supabase + cache
lib/booking/shopExtrasStorage.ts	Owner profile, schedule, services, offers
lib/booking/shopSchedule.ts	Weekly hours, slot builder, open-now
lib/booking/reporting.ts	Date ranges, EGP, buildOwnerReportHtml()
lib/booking/commerceEvents.ts	Owner/customer notifications
lib/booking/bookingReminders.ts	60/30 min reminders (native)
lib/booking/partsStorage.ts	Store inventory + orders
lib/booking/wash/washBranchStorage.ts	Multi-branch state (Supabase-first + cache)
lib/booking/wash/branchRepository.ts	branch_employees CRUD
lib/booking/wash/branchManagerRepository.ts	Manager account RPC + signUp flow
lib/shop/ownerRegistrationRepository.ts	Merchant signUp + register_shop_owner RPC
lib/admin/adminRepository.ts	Pending list, merchants, stats, approve/reject
components/admin/AdminPanel.tsx	Super Admin UI (3 tabs)
lib/shop/shopStaffUser.ts	Resolve owner/manager/admin/pending from users
lib/supabase/client.ts	Supabase singleton (env → app.json → fallback)
lib/push/shopPush.ts	Expo push for owners (native)
lib/i18n/strings.ts	~928 translation keys × EN + AR
lib/ui/userAlert.ts	window.alert on web (Alert.alert fallback)
lib/ai-chat.ts	OpenAI assistant

11. Internationalization

العربية والإنجليزية

- Locales: en | ar — stored in @pitstop/locale
- Default from device language (expo-localization)
- API: `t(key)`, `tp(key, vars)`, `isRTL`
- RTL: `I18nManager` (native) + `document.dir` (web)
- Toggle: Welcome screen + Settings

12. Notifications & PDF Reports

الإشعارات وتقارير PDF

Owner push (native only)

- Table: shop_push_tokens
- Triggers: new booking, new parts order
- Via Expo Push API

Customer in-app notifications

Kinds: booking_approved, booking_declined, booking_reminder, parts_order_confirmed, parts_order_cancelled

PDF reports

- `buildOwnerReportHtml()` — shop header, date range, gross/fee/net
- Web: scrollable modal preview → Save as PDF (browser print)
- Native: `printToFileAsync` + share
- Used in generic shop panel + WashOwnerPanel

13. Demo Accounts

demo123 — كلمة السر: demo123

Platform admin

admin@demo.com / demo123 — set role manually after step5 SQL:

```
update public.users set role = 'admin', is_active = true, shop_id = null
where email = 'admin@demo.com';
```

Shop owners (pre-approved in seed)

Email	Shop ID	Type
wash@demo.com	shop-wash-nile	wash
maintenance@demo.com	shop-maint-autofix	maintenance
parts@demo.com	shop-parts-nasr	parts
accessories@demo.com	shop-accessories-maadi	accessories
winch@demo.com	shop-winch-maadi	winch

Also: wash2@, wash3@, maintenance2@, maintenance3@, parts2@, accessories2@, winch2@ — see seed.sql

Branch manager

manager.wash@demo.com / demo123 — Nile Auto Wash main branch

Customers

customer@demo.com · sara@demo.com / demo123

14. Environment & Running the App

تشغيل المشروع

Environment variables (.env)

```
EXPO_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=your-anon-key
EXPO_PUBLIC_OPENAI_API_KEY=sk-... # for AI assistant
```

Fallback (app.json extra)

Supabase URL and anon key also in app.json → extra if .env unset.

Commands

```
cd car-care-eg
npm install
npm start          # Expo (offline mode)
npm run web        # http://localhost:8082
npm run android
npm run ios
```

Supabase project (demo)

URL: <https://qlopvpyeawauepsyrirz.supabase.co>

15. Limitations & Roadmap

القيود والتطوير المستقبلي

Current limitations

Limitation	Detail
Hybrid storage	Many features use AsyncStorage; Supabase when configured
Guest cannot book	Login required despite guest browsing
Merchant approval	New owners need admin approve before shop goes live
Winch not on home	In DB but no home service card
Web push disabled	Owner push + reminders native only
Payments	No online payment integration yet
Legacy garage	Parallel SQLite app (hidden tabs)

Future enhancements (from docs)

- Online payments (Paymob/Fawry)
- Loyalty program & subscriptions
- Driver network on Supabase + moderation
- Vehicle cloud sync
- Platform admin dashboards
- Multi-region (GCC expansion)
- Assign employees to bookings/shifts
- QR check-in, queue tracking, fleet management

Related documents in repo

- docs/SCHEMA.md — database reference
- docs/FUTURE_ENHANCEMENTS.md — product roadmap
- docs/pitstop-2.0-owner-srs.html — Owner/Branch SRS
- pitstop2.0.pdf — Customer-side SRS (original)

16. UI Architecture & Design System

بنية الواجهة ونظام التصميم

Layout model

```
app/_layout.tsx (Root Stack)
├─ welcome, auth-required, book, shop-profile, nearby, parts-shop ...
└─ (tabs)/_layout.tsx (Bottom tabs – customer OR shop mode)
    ├─ index (Home)
    ├─ bookings | favorites | assistant | settings ← customer area
    └─ shop ← owner dashboard (tabs hidden when shop logged in)
```

Two UI modes

Mode	Trigger	Tabs shown	Default route
Customer / Guest	No shop session	Home, Bookings, Favorites*, Assistant, Settings	/ (Home)
Shop staff	ShopAuthContext.shop set	Shop tab only (bar hidden)	/shop → AppBootstrap forces /shop

*Favorites tab hidden for guests — login required.

Theme (constants/Theme.ts + ThemePreferenceContext)

- **Light / Dark** — user choice in Settings; persisted
- **Tokens:** bg, bgElevated, card, border, text, textMuted, accent, warm, onAccent
- **RTL:** Arabic flips layout via I18nContext + I18nManager
- **Owner wash accent:** warm color in dark mode for owner buttons

UI patterns used everywhere

Pattern	Where	Behavior
Pressable + gradient btn	welcome.tsx	Login/register; pointerEvents none on gradient (web fix)
OwnerSectionCard	WashOwnerPanel	Collapsible section with title + subtitle
Modal confirm	shop.tsx, WashOwnerPanel	Replaces Alert.alert on web for booking decisions
userAlert	welcome, book, shop	window.alert on web when Alert invisible
ScrollView sections	All owner panels	Long vertical CMS layout
iframe report preview	shop.tsx, WashOwnerPanel	Scrollable HTML report before PDF

17. Master Navigation Flowcharts

مخططات التنقل الرئيسية

17.1 App entry & auth routing

```
[App launch]
|
v
[AppBootstrap waits: CustomerAuth + ShopAuth ready]
|
+-- isAdmin? -----> router.replace('/admin')
|
+-- isPendingOwner? -----> /welcome?pending=1 (blocked from /shop)
|
+-- approved shop session? --> router.replace('/shop')
|
+-- no session + protected? --> /welcome
|
+-- customer/guest on /welcome? --> / or returnTo
|
v
[/ (tabs) or stack screens]
```

17.2 Customer booking journey

```
Home (/ (tabs)/index)
| tap service card (maintenance|wash|parts|accessories)
v
/service/[type] → pick area (Maadi, Nasr City...)
v
/service/[type]/[areaId] → shop list
|
+-- parts/accessories --> /parts-shop/[shopId]
|
+-- maintenance/wash --> /shop-profile/[shopId]
|
| [Book button]
v
guest? --> /auth-required --> /welcome?returnTo=/book/...
|
v
/book/[shopId]?serviceId=...
| fill phone, car, date, time slot, notes
v
createBooking() --> success modal --> /bookings tab
```

17.3 Alternative discovery paths

```
Home --search/filter--> service cards
Home --live offers--> shop-profile
Nearby GPS (/nearby/[type]) --> shop-profile | parts-shop
Favorites tab --> shop-profile (login required)
Shop profile: Call | Directions | Share | Favorite | Gallery | Reviews
```

17.4 Owner login & panel routing

```
/welcome → Owner tab
|
```

```

+-- Login (approved owner/manager)
|   v
|   ShopAuthContext.login() → ok
|   v
|   router.replace('/shop') → WashOwnerPanel | store UI | generic UI
|
+-- Login (admin)
|   v
|   router.replace('/admin') → AdminPanel
|
+-- Login (pending_owner)
|   v
|   pending_approval message – no shop access
|
+-- Register as merchant
|   v
|   registerOwner() → signUp + register_shop_owner RPC
|   v
|   pending_owner + shop is_active=false → wait for admin

```

17.5 Wash owner hub (secondary navigation)

```

WashOwnerPanel → Hub shortcuts
  v
/shop/wash-owner-hub?tab=notifications|orders|history
|
+-- notifications: wash notification inbox
+-- orders: approve/reject/in_progress/done bookings
+-- history: completed/cancelled list + clear

```

17.6 Branch manager vs owner (wash UI)

```

Same login form (/welcome Owner tab)
  v
ShopAuthContext resolves role
|
+-- isOwner --> branch tabs visible, create branch, create manager
|
+-- isBranchManager --> single branch, no manager section, bookings filtered by branch_id

```

18. Customer UI — Module by Module

واجهة العميل — شرح كل جزء

Module A — Welcome / Auth (app/welcome.tsx)

UI elements: Logo, language toggle, Customer | Owner mode tabs, guest button, form box.

Customer tab: Login ↔ Register toggle, email/password, optional vehicles on register, forgot password link.

Owner tab: Shop email + password, demo accounts hint.

Actions: onCustomerSubmit → login/register; onOwnerSubmit → shop login; continueAsGuest → home without Supabase.

Feedback: formMessage inline (red) + userAlert on errors; success → router.replace.

Module B — Home (app/(tabs)/index.tsx)

Sections top→bottom:

1. Header logo + sign out (if logged in)
2. Car profile card — saved make/model; edit modal
3. Next booking card — upcoming appointment or empty state
4. Live offers carousel — wash/maintenance promotions
5. Service search + filter (all / wash only)
6. 4 service cards → navigate to /service/{type}

Module C — Area & Shop List (app/service/)

Step 1: Grid of Cairo/Giza areas with shop count.

Step 2: Shops in selected area; card shows name, rating, distance hint, open status.

Routing: parts/accessories → parts-shop; others → shop-profile.

Module D — Shop Profile (app/shop-profile/[shopId].tsx)

UI block	User action
Hero image + gallery swipe	View photos
Name, address, phone	Call, open maps
Schedule line	See open hours
Services list	Tap service → book with serviceId
Offers	Read promotions
More info	Owner free-text details
Reviews	Read/submit rating (logged in)
Share / Favorite	Social share; save shop
Book CTA (sticky)	→ /book or auth-required

Module E — Booking Form (app/book/[shopId].tsx)

Field	Source	Validation
Phone	Customer profile	Egypt E.164
Car type / color	Vehicle storage or manual	Car type required
Date picker	BookingDatePicker	Owner schedule days
Time slots grid	buildSlotsForShopDate	Required; booked slots disabled
Service summary	Selected service or shop default	Price + duration shown
Notes	Free text	Optional
Submit	createBooking	Disabled if no slot

Success: Modal → View my bookings → /bookings tab.

Module F — My Bookings (app/(tabs)/bookings.tsx)

List cards: shop name, service, datetime, status badge, cancel button (pending/confirmed).

Clear history action for done/cancelled items.

Module G — Favorites, Assistant, Settings

Favorites: Saved shop cards → shop-profile. Login required.

Assistant: Driver Network posts (local) + AI chat panel (OpenAI key required).

Settings: Language EN/AR, theme light/dark, vehicles link, support links, privacy, sign out.

Vehicles: /settings/vehicles — add/edit/delete customer cars.

Module H — Parts Shop (app/parts-shop/[shopId].tsx)

Product grid, cart badge, qty +/-, shipping address, phone, checkout → createPartsOrder.

No time slots — inventory-based e-commerce flow.

Module I — Nearby (app/nearby/[type].tsx)

Requests GPS permission → sorts shops by distance.

Filter chips: All, Top rated, Price, Distance, Open now, Favorites.

Module J — Customer Notifications (CustomerNotificationsBell)

Bell icon in tab headers (logged-in customer only).

Shows booking approved/declined/reminder + parts order updates.

Processes due reminders on focus.

19. Owner UI — Module by Module

واجهة صاحب المحل — شرح كل جزء

19.1 Generic owner panel (app/(tabs)/shop.tsx — maintenance / winch)

Header bar

Profile cover image, notifications bell, logout. Shop name from catalog.

Profile section

Fields: profile name EN/AR, address EN/AR, phone, email, more info EN/AR.

Maintenance only: Winch service toggle (shows winch add-on to customers).

Save profile info button → shopExtrasStorage.

Photo album

Add image URL, remove image. Gallery shown on shop-profile.

Service price + schedule

Default price EGP, open/close time, slot duration (15+ min). Friday closed default.

Generates customer time slots on book screen.

Offers

Add bilingual offer title/body, expiry. Cancel active offers.

Bookings dashboard

Report: date from/to, quick last N days, preview list, View report modal, Save PDF.

Active requests: pending + confirmed — Confirm | Cancel | Mark done buttons via confirm modal.

History: done + cancelled — show/hide toggle.

19.2 Store owner panel (parts / accessories)

Same profile + gallery as generic. **No schedule/bookings.**

Inventory table: add item (name, price, stock, image URL), +/- stock buttons.

Orders table: customer phone, items, total, status — Confirm | Cancel | Shipped.

19.3 Wash owner panel (WashOwnerPanel.tsx) — section order

#	Section (i18n key)	UI controls	Owner only?
1	wash_dashboard_*	Stat cards: today bookings, pending, revenue, rating, customers, top service	Stats scope differs
2	wash_status_*	Open / Closed / Busy / Vacation + vacation dates	No
3	wash_analytics_*	Revenue & booking charts (computed locally)	No

4	wash_hub_*	3 shortcut buttons → wash-owner-hub tabs	No
5	wash_branch_*	Branch tabs, add branch, rename	Owner only
6	wash_profile_*	Branch profile fields + save	No
7	wash_gallery_*	Image URL list add/remove	No
8	wash_base_price_*	Default service price input	No
9	wash_services_*	Service list, add/edit modal (EN/AR, price, duration, category, visible, sort)	No
10	wash_hours_*	Day picker 0–6, open/close/break, copy to all days	No
11	wash_coupons_*	Coupon list, add percent/fixed code	No
12	wash_manager_*	Link email Create login (name, email, password)	Owner only
13	wash_employees_*	Employee list per branch, add/edit/delete roster	No
14	wash_reviews_*	Hide review, reply, report	No
15	wash_report_*	Date range report + PDF (same engine as generic)	No
16	Active bookings inline	Status workflow: accept, reject+note, in progress, done, no show	Branch filtered

19.4 Wash owner hub screens

wash-owner-hub.tsx

Tab: notifications → list from washNotificationCenter (unread badge)

Tab: orders → booking cards + action buttons + call customer

Tab: history → past bookings + delete/clear

19.5 Super Admin panel (AdminPanel.tsx)

Tab	Module	Actions
Dashboard	Platform stats cards	Read-only metrics from bookings + users
Pending	PendingOwnerRequest list	Accept / Reject → Supabase RPC
Merchants	ActiveMerchant list	View approved owners + branch count

Entry: /welcome Owner tab → admin login → /admin. Uses OwnerSectionCard for sections.

20. Screen-by-Screen Reference

مرجع سریع لكل شاشة

/welcome Auth gateway. Customer register/login, guest, owner login. Language toggle.	/(tabs)/index Home hub. Services, offers, next booking, car profile.
/service/[type] Pick geographic area.	/service/[type]/[areald] Shop list for area.
/shop-profile/[shopId] Public shop page before booking.	/book/[shopId] Appointment form + slot picker.
/bookings tab Customer appointment list.	/favorites tab Saved shops (login).
/assistant tab Driver network + AI chat.	/settings tab App preferences.
/settings/vehicles Customer car list CRUD.	/nearby/[type] GPS shop discovery.
/parts-shop/[shopId] Storefront + cart checkout.	/auth-required Login gate with returnTo.
/(tabs)/shop Owner dashboard (type-specific panel).	/shop/wash-owner-hub Wash notifications/orders/history.
/admin Super Admin: dashboard, pending merchants, active merchants.	/reset-password Supabase password reset handler.

Tab visibility matrix

Tab	Guest	Customer	Shop owner
Home	✓	✓	hidden
Bookings	✓	✓	hidden
Favorites	hidden	✓	hidden
Assistant	✓	✓	hidden
Settings	✓	✓	hidden
Shop	hidden*	hidden	✓ (only tab)

*Shop tab visible when no customer area and not logged in as customer — rare edge case.

21. Shared UI Components

المكونات المشتركة

Component	Path	Used for
AppBootstrap	components/AppBootstrap.tsx	Auth routing, splash hide
OwnerSectionCard	components/owner/OwnerSectionCard.tsx	Wash owner CMS sections
ServiceOptionCard	components/ui/ServiceOptionCard.tsx	Home service type cards
BookingDatePicker	components/ui/BookingDatePicker.tsx	Book screen date selection
CustomerNotificationsBell	components/customer/CustomerNotificationsBell.tsx	In-app notification dropdown
ShopCatalogProvider	context/ShopCatalogContext.tsx	Loads areas/shops cache for UI lists
userAlert	lib/ui/userAlert.ts	Cross-platform error dialogs

Context → UI binding

Context	UI reads	UI effect
I18nContext	t(), locale, isRTL	All labels; remount tabs on locale change
ThemePreferenceContext	useAppTheme()	Colors on every screen
CustomerAuthContext	customer, isGuest, busy	Auth gates, profile prefill
ShopAuthContext	shop, isOwner, isBranchManager	Panel type, branch tabs, manager section
ShopCatalogContext	ready, version	Shop lists, home offers

22. UI State & Context Flow

تدفق الحالة بين الواجهة والبيانات

22.1 Customer books — UI ↔ data

```
[Book screen state: phone, carType, dateYmd, timeSlot, notes]
  |
  | onSubmit
  v
createBooking() → Supabase bookings + AsyncStorage cache
  |
  +→ pushOwnerNotification (owner bell)
  +→ pushWashCenterNotification (wash hub)
  +→ sendShopPushForBooking (native push)
  |
  v
[Success modal] → user taps → /bookings tab reloads listBookingsForPhone()
```

22.2 Owner confirms — UI ↔ data

```
[shop.tsx or wash-owner-hub booking card]
  |
  | Confirm button → confirm modal
  v
updateBookingStatus(id, 'confirmed')
  |
  +→ pushCustomerNotification('booking_approved')
  +→ scheduleBookingReminders (60m, 30m)
  |
  v
[CustomerNotificationsBell shows approval]
[Bookings tab status badge updates on refresh]
```

22.3 Wash branch switch — UI ↔ data

```
[Owner taps branch tab in WashOwnerPanel]
  |
  v
setActiveBranch(branch) + useEffect reload
  |
  +→ loadBranchEmployees(branch.id) from Supabase
  +→ applyBranchToForms() fills profile/gallery/hours fields
  |
  v
[All section forms show selected branch data]
[Save writes back to washBranchStorage → branchRepository Supabase sync]
```

22.4 Report PDF — UI ↔ data

```
[Owner sets date range + Apply]
  |
  v
filterBookingsByDateRange() from listBookingsForShop()
  |
  v
buildOwnerReportHtml() → HTML string
  |
```

+— Web: setReportPreviewHtml → Modal iframe (scroll) → Save PDF = print
+— Native: Print.printToFileAsync → Sharing.shareAsync

22.5 Local vs cloud (what UI shows when offline)

UI area	Offline behavior
Shop lists	Cached catalog from last Supabase refresh
Bookings list	AsyncStorage merge; may differ from cloud
Owner extras (schedule, offers)	Local shopExtrasStorage until synced
Wash branches	Local cache + Supabase when online
Login	Requires Supabase (shows not_configured if missing)
Admin panel	Requires Supabase + step5 RPCs; read-only offline

23. Super Admin — Overview & Roles

لوحة Super Admin — نظرة عامة

PitStop 2.0 adds a **merchant approval gate**: new shop owners cannot go live until a platform admin approves them. Existing seed owners (e.g. wash@demo.com) remain pre-approved.

State machine — merchant account

State	users.role	users.is_active	shops.is_active	Customer catalog	App route
New registration	pending_owner	false	false	Hidden	/welcome (pending message)
Approved	owner	true	true	Visible	/shop
Rejected	customer	true	deleted	—	/ (customer area)

Admin capabilities

- View platform-wide booking stats and 12% platform fee estimate
- List pending merchant registration requests
- Accept → atomic RPC activates user + shop
- Reject → deletes shop row, resets user to customer
- List active merchants with branch counts

24. Super Admin — Flowcharts

مخططات Super Admin والتسجيل

24.1 Merchant self-registration

```
/welcome → Owner tab → "Register as merchant"
|
v
Fill: business name, owner name, phone, address, shop type, area, email, password
|
v
ShopAuthContext.registerOwner()
|
+-- supabase.auth.signUp({ email, password, metadata })
|
+-- RPC register_shop_owner(...)
|   |
|   +-- INSERT shops (is_active=false, owner_user_id)
|   +-- INSERT shop_branches (slug=main)
|   +-- UPDATE users SET role=pending_owner, is_active=false, shop_id
|
v
Show: "Pending approval" (owner cannot access /shop yet)
|
v
[Admin reviews in /admin → Pending tab]
```

24.2 Admin approval flow

```
Admin logs in (admin@demo.com) via Owner tab on /welcome
|
v
AppBootstrap → router.replace('/admin')
|
v
AdminPanel → Pending requests tab
|
+-- [Accept] → RPC approve_shop_owner(user_id, shop_id)
|   |
|   +-- users: role=owner, is_active=true
|   +-- shops: is_active=true
|   |
|   v
|   Shop appears in customer catalog (refreshCatalog)
|   Owner can login → /shop
|
+-- [Reject] → RPC reject_shop_owner(user_id, shop_id)
|   |
|   +-- DELETE shop_branches + shops
|   +-- users: role=customer, shop_id=null
|   |
|   v
|   Registration removed; user stays as customer
```

24.3 Catalog visibility

```
refreshCatalog() queries shops WHERE is_active = true
|
```

```

  v
pending_owner shops (is_active=false) NEVER shown to customers
|
  v
After approve_shop_owner → shop included on next catalog refresh

```

24.4 Platform stats calculation

```

AdminPanel → Dashboard tab
|
  v
fetchPlatformStats() in adminRepository.ts
|
+-- COUNT all bookings
+-- FILTER status = 'done' for revenue
+-- SUM platform_fee_egg OR 12% of service_price_egg
+-- COUNT pending_owner users
+-- COUNT active owner users
|
  v
Display stat cards (OwnerSectionCard layout)

```

25. Super Admin — Modules & UI

شرح وحدات لوحة المشرف

Route: `app/admin/index.tsx` + `components/admin/AdminPanel.tsx`

Full-screen layout (web-optimized): fixed header + horizontal tab chips + scrollable content.

Tab A — Dashboard (`admin_tab_dashboard`)

Stat card	Data source
Total bookings	COUNT bookings
Completed	status = done
Gross revenue	SUM service_price_egp (done)
Platform fee (12%)	SUM platform_fee_egp or calculated
Pending owners	COUNT users role=pending_owner
Active merchants	COUNT users role=owner, is_active

Tab B — Pending requests (`admin_tab_pending`)

Each card shows: shop name, type, owner email, phone, address.

Accept → `approveShopOwner()` → RPC

Reject → `rejectShopOwner()` → RPC

Uses `userAlert` for success/error on web.

Tab C — Active merchants (`admin_tab_merchants`)

Lists approved owners: shop name, type, email, active branch count from `shop_branches`.

Header actions

- Title: Super Admin / لوحة المشرف
- Signed-in email from `ShopAuthContext.staff`
- Sign out → `useAppSignOut()` → `/welcome`

26. Merchant Registration Module

وحدة تسجيل التاجر الجديد

UI: `app/welcome.tsx` — **Owner tab**

Toggle: Login ↔ Register as merchant (`owner_register_link`)

Register fields:

1. Business name (`shopName`)
2. Owner full name (`name`)
3. Egypt mobile phone — validated
4. Business address
5. Shop type chips: wash, maintenance, parts, accessories
6. Area chips: from catalog areas (Maadi, Nasr City, ...)
7. Email + password (strong password rules)

Submit: `registerOwner()` in `ShopAuthContext`

Backend: `lib/shop/ownerRegistrationRepository.ts`

1. `supabase.auth.signUp` with metadata (`name`, `phone`, `account_type=shop_owner`)
2. `supabase.rpc('register_shop_owner', { p_shop_name, p_shop_type, p_area_id, ... })`
3. On RPC failure → `signOut` and return invalid
4. On success → session restored as `pending_owner`

Login while pending

Credentials valid but `login()` returns `pending_approval`.

AppBootstrap keeps user on `/welcome` with pending message — no access to `/shop` or customer booking as owner.

27. New Library Modules (Admin)

ملفات الكود الجديدة

File	Exports / purpose
lib/admin/adminRepository.ts	listPendingOwnerRequests, listActiveMerchants, fetchPlatformStats, approveShopOwner, rejectShopOwner
lib/shop/ownerRegistrationRepository.ts	registerShopOwner(input) → signUp + RPC
lib/shop/shopStaffUser.ts	fetchAppStaffUser, resolveShopSession — extended for admin + pending_owner
context/ShopAuthContext.tsx	isAdmin, isPendingOwner, registerOwner, pending_approval login result
components/AppBootstrap.tsx	Admin → /admin; pending → /welcome?pending=1
components/admin/AdminPanel.tsx	Full Super Admin UI
app/admin/_layout.tsx	Stack layout for admin route
supabase/apply-pitstop-2.0-step5-admin-approval.sql	Enum, RLS, RPCs, is_platform_admin()

Context flags quick reference

Flag	When true	Route
isAdmin	users.role = admin	/admin
isPendingOwner	users.role = pending_owner	/welcome (blocked)
isOwner	users.role = owner, active	/shop
isBranchManager	users.role = branch_manager	/shop (branch filtered)